Geological Modelling Report:

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Course: CIV185
Date: Apr 4

1. Introduction

This report details the geological modelling process for Tutorial 3 of CIV185, utilizing the Visible Geology tool to reconstruct a hypothetical geological scenario. As the assignment instructions require, the model includes key geological structures such as an unconformity, sedimentary layers, a normal fault, and a valley.

2. Model Construction and Geological History

2.0 Rock Formations and Assignments

Below is a list of the formations and their corresponding letter assignments as per the provided model:

Formation A: Limestone (~60m thick)
Formation B: Sandstone (~50m thick)
Formation C: Shale (~40m thick)
Formation E: Conglomerate (~60m thick)
Formation F: Basalt (~30m thick)

These formations are critical in understanding the geological history and the impact of structural deformations.

2.1 Initial Deposition

The base layer represents a pre-existing rock formation that underwent erosion, forming an unconformity. This indicates a significant time gap in the geological record. Over this surface, two sedimentary layers were deposited:

Layer 1: Sandstone (~50m thick)
Layer 2: Shale (~40m thick)

These layers suggest a shift from a high-energy depositional environment (sandstone) to a low-energy, deep-water setting (shale).

2.2 Fault Formation

A normal fault was added with a slip of 15m and a strike/dip of 035/35°E, indicating extensional tectonic forces. The fault cuts through all layers, displacing them and creating a structural discontinuity.

2.3 Valley Formation & Topography Effects

A valley was introduced to simulate surface erosion. This process exposes different geological layers at the surface, which then also leads to different layer thickness where there's an intersection of the topography, altering their original continuity. The valley also modifies the cross-sectional view by truncating certain units.

3. Cross-Sectional Views

3.1 Strike-Perpendicular Cross-Section

Figure 1 below shows the perpendicular cross-section. The fault offsets the stratigraphic units, with the downthrown block shifting the upper layers downward and revealing the true thickness of each layer.

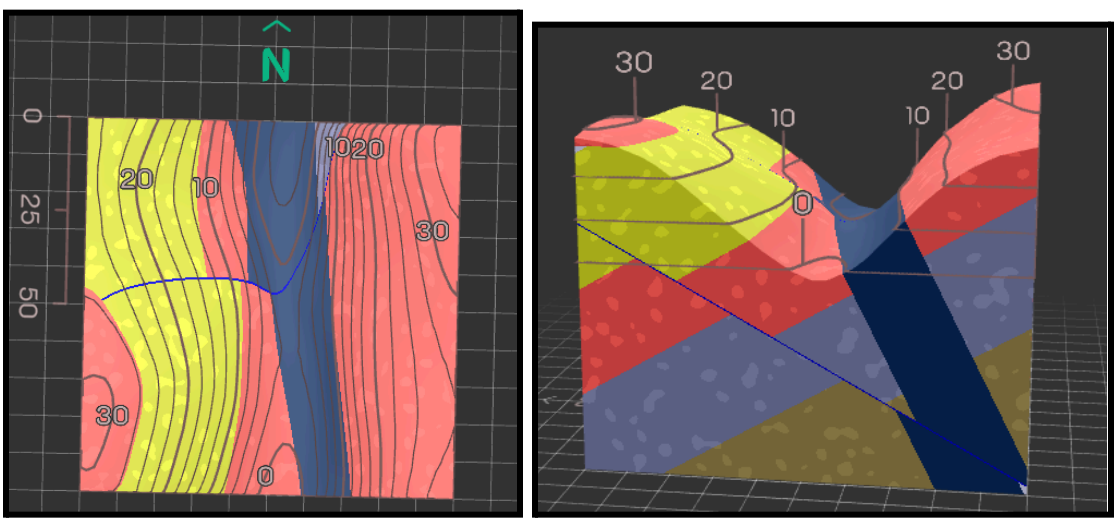

Figure 1. Strike-perpendicular cross section

3.2 Strike-Parallel Cross Section

Figure 2 shows the strike-parallel cross-section. This view provides insight into lateral continuity and the degree of displacement caused by faulting and the dip directions of the layers.

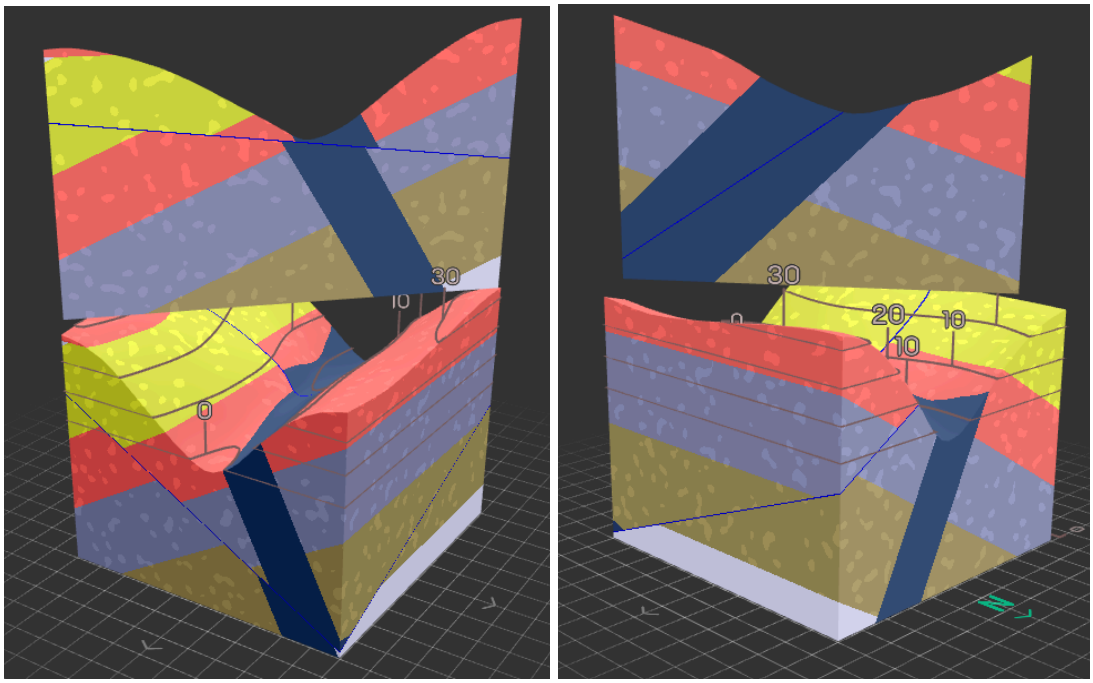

Figure 2. Strike-parallel cross section

4. Drill Cores Analysis

Two drill cores were added to the model:

Vertical Drill Core: Provides an undisturbed sequence of rock layers.
Inclined Drill Core: Intersects the fault, confirming the displacement observed in cross-sections.

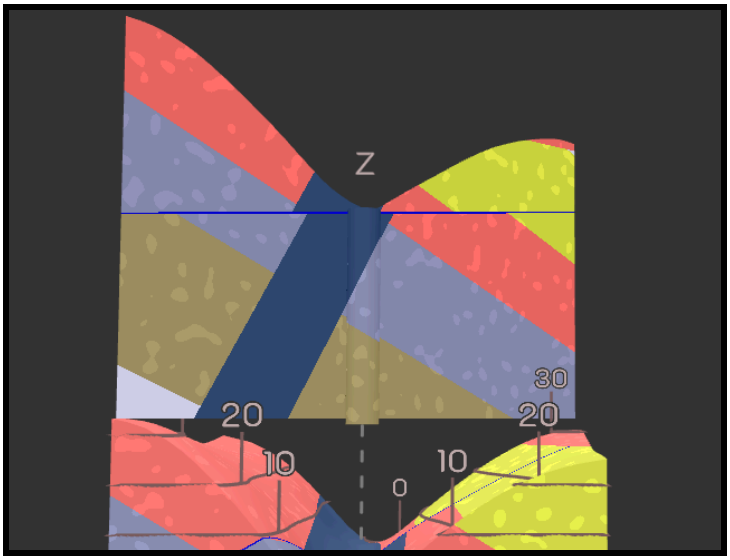

Figure 3. Vertical Drill Core

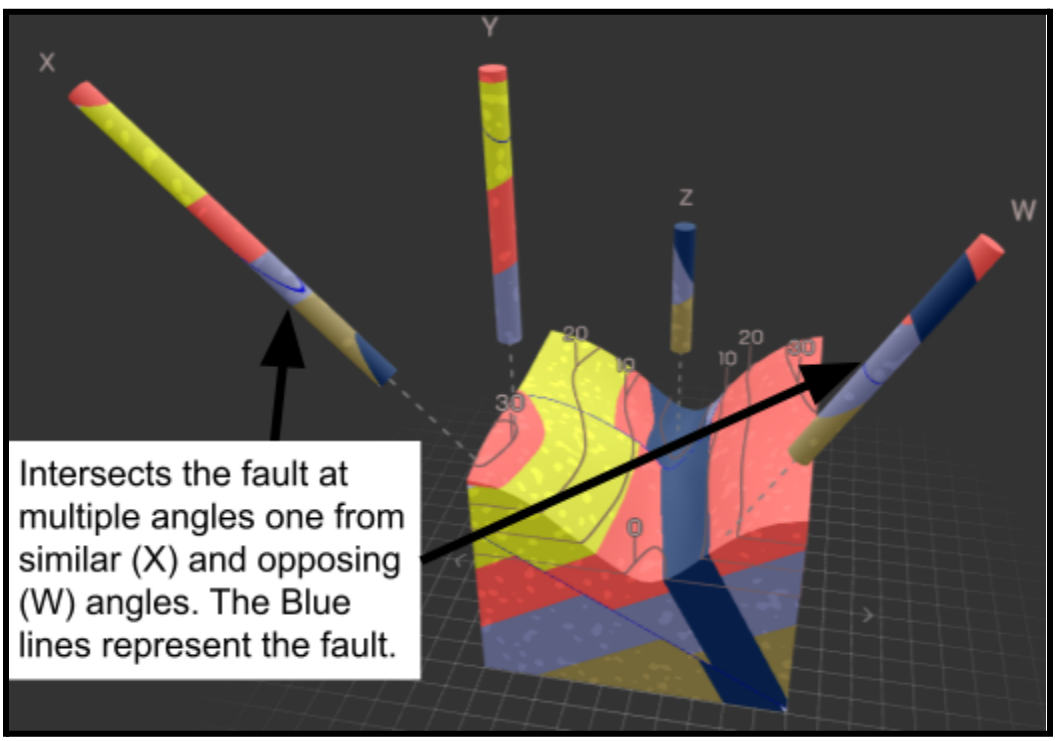

Figure 4. Inclined Drill Core

5. Conclusion

This geological model successfully reconstructs a sedimentary sequence, fault displacement, and erosion processes. The valley significantly alters surface exposure, while the fault introduces an apparent structural deformation. These elements highlight fundamental geological principles applied in structural geology and terrain evolution.